

Computer Networks

CSIT — 4th Semester — 2082 — Answer Sheet
Subject Code: CSC263

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What are the advantages of layered architecture in network communication? Explain TCP/IP protocol stack in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What are the services provided by Transport Layer? Explain working principle of Distance Vector Routing Protocol.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Differentiate between IPv4 and IPv6 Protocol in brief. Explain Cyclic Redundancy Check with suitable numerical example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. How connection is established in TCP protocol using 3-way handshaking? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Write subnet ID, broadcast address and address range of each subnet if you divide a class C network 193.168.2.0/24 in 8 different subnets. What is the new subnet mask?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is the major purpose of DNS? Explain how DNS resolve domain name to IP address.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Discuss different guided transmission medium in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Briefly explain Go-Back-N ARQ and Selective Reject ARQ protocols.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is HTTP protocol? Explain HTTP Request and HTTP Response message in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is virtual circuit network? Explain ATM as a virtual circuit wide area network.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is software defined network? Explain in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes: a. Leaky Bucket b. FTP Protocol

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.